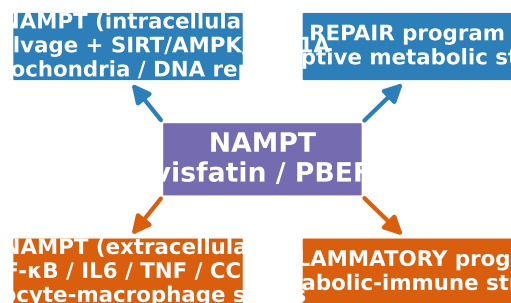


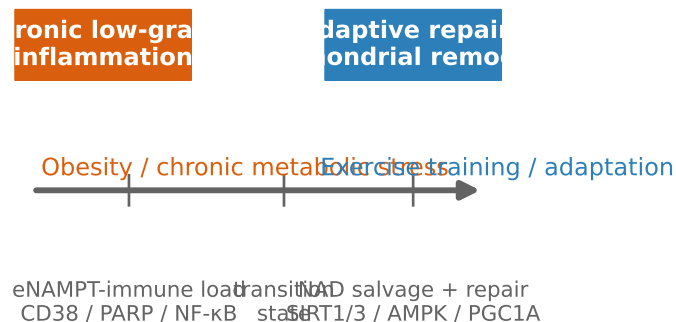
# State-dependent NAMPT-NAD inflammatory-repair axis in obesity and exercise adaptation

## a NAMPT dual identity



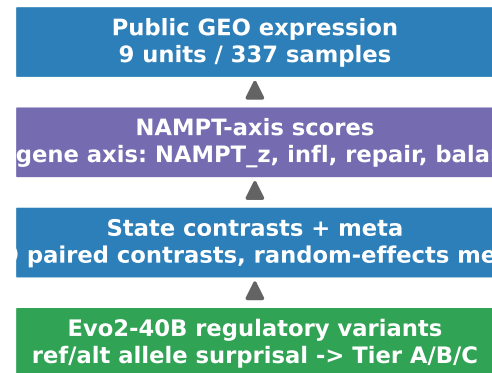
Key question: when does NAMPT signal repair vs. inflammatory load?

## b State-dependent axis



The same gene set, different states  
-> balance\_score flips direction

## c Multi-layer public-data evidence



## d Repair program (transcript evidence)

co-expression support:  
GSE272133 repair modules  $\alpha = 0.79-0.91$   
GSE305038  $\alpha$  mean 0.78; GSE32575  $\alpha$  mean 0.61

NAMPT NMNAT1/2/3 NAPRT NADSYN1  
SIRT1 SIRT3 SIRT6 PRKAA1/2  
PPARGC1A TFAM NRF1 SOD2  
CAT GPX1 ATG5 BECN1 XRCC1

## e Inflammatory program (transcript evidence)

state evidence:  
acute exercise: NAMPT\_z +0.85 (92% agree)  
balance -0.62 (83%); NF-kB-module sensitive

NAMPT CD38 BST1 PARP1/2  
NFKB1 RELA TNF IL6 IL1B  
CCL2 CXCL8 TLR4 NLRP3  
CASP1 CD14 CD68 ITGAM

Acute exercise (k=3) 0.12 †  
Obesity (k=3) 0.10 †  
Cell model (k=2) 0.07 †  
6-wk training (n=3) 0.27 †

repair-negative <- balance -> repair-positive  
† directional summary (limited k/n); full CIs in Fig. 4-5